MILITARY SPECIFICATION

NEON DRIVER CIRCUIT BOARD: 8623298

1. SCOPE

1.1 This specification covers one type of neon driver circuit board, part of the Test Set, Computer Logic Unit AN/GSM-70.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18, Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army Munitions Command

F8623298	Neon Driver Assembly
F8623299	Schematic, Electrical, Neon Driver

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

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(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The neon driver circuit board shall be manufactured in accordance with Drawing F8623298 and drawings pertaining thereto.

3.2 General specifications.- The circuit board shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The circuit board shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the circuit board for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the circuit board shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The circuit board shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to temperatures of -40°F and +135°F.

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3.3.3.2 Storage.- The circuit board shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The circuit board shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The circuit board shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The circuit board shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Electrical performance.- The circuitry shall be in accordance with electrical schematic, Drawing F8623299. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.1 Neon driver.- The following requirements shall pertain to each of the 12 neon driver circuits.

3.4.1.1 True state - (output loaded).- With a true signal (minus 6 plus or minus 0.18 volts) applied to the input gate, and the load of Figure 1 connected to the output, the following conditions shall exist within a driver:

(a) The emitter voltage shall be minus 3.7 plus or minus 0.22 volts.
(b) The collector voltage shall be 40 plus or minus 10 volts.

3.4.1.2 True state - (output not loaded).- With a true input to the driver, the voltage between the collector and the plus 150 volts shall be no less than 135 volts.

3.4.1.3 False state - (output not loaded).- With a false condition (minus 0.15 plus or minus 0.15 volts) applied to the input, the following conditions shall exist within a driver:

(a) The emitter voltage shall be no more negative than minus 1.1 volts.
(b) The collector voltage shall be plus 100 plus or minus 10 volts.
(c) The requirement of condition (a) above shall apply when any minus 6 volts ground combination is used at the input gate.

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3.4.2 Threshold amplifier.- Each of the 3 amplifiers shall meet the following requirements:

3.4.2.1 Input diodes.- The input diodes shall have:

(a) Leakage current of no more than 0.15 microamps when reverse biased by 35 volts.
(b) The forward voltage drop across the diodes shall not exceed 0.7 volt at 22 plus or minus 1.1 microamps.

3.4.2.2 Static conditions - (output and output prime loaded).- With separate loads indentical to figure 2 connected to both the output and output prime of the threshold amplifier, and the input network of figure 3 connected to the base of the input transistor, the following conditions shall exist within the amplifier:

(a) With the input adjusted to supply minus 1.67 plus or minus 0.02 volts:

(1) The voltage at the output shall be between 0 volts and minus 0.25 volts.
(2) The voltage at the output prime shall be minus 6.8 plus or minus 0.5 volts.

(b) With the input adjusted to supply minus 1.83 plus or minus 0.02 volts:

(1) The voltage at the output shall be minus 6.8 plus or minus 0.5 volts.
(2) The voltage at the output prime shall be between 0 volt and minus 0.25 volts.

3.4.3 Input network.

3.4.3.1 The forward voltage drop across each of the input network diodes shall be no greater than 0.6 volts when carrying a current of 6 plus or minus 0.3 milliamperes.

3.4.3.2 The reverse leakage current of each of the input network diodes shall be no greater than 6 microamperes with 6 plus or minus 0.5 volts applied to the diode in the reverse direction.

3.4.4 ISR-OSR drivers.- Under the input and load conditions of figure 4, the drivers shall perform as follows:

3.4.4.1 True state.- With a true input applied to the base of the input transistor, the following conditions shall exist within a driver:

(a) The collector voltage of the input transistor shall be a minus .05 plus or minus .05 volts.
(b) The voltage at the base of the output transistor shall be a minus 5.65 plus or minus 0.54 volts.

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(c) The collector voltage of the output transistor shall be a minus 5.8 plus or minus 0.54 volts.

3.4.4.2 False state.- With a false input applied to the base of the input transistor, the following conditions shall exist within a driver:

(a) The collector voltage of the input transistor shall be a minus 18 plus or minus 0.6 volts.
(b) The collector voltage of the output transistor shall be a plus 2.93 plus or minus 0.55 volts.

3.5 Temperature extremes.- The following requirements shall be met at the conditions of 3.3.3.1, and 3.3.4.1 with the input voltages compensated as indicated in 4.4.1.1.

3.5.1 Neon driver.

3.5.1.1 OFF state - (no load).- With a false input to the drivers (any true-false combination input), the voltage at the collector with respect to the 150 volt supply shall not exceed 70 volts.

3.5.1.2 ON state - (no load).- With a true input to the drivers, the voltage at the collector with respect to the 150 volt supply shall be no less than 110 volts.

3.5.1.3 ON state - (output loaded).- With a 56K ohm load resistor at the output, the voltage across this resistor, with true inputs, shall be no less than 100 volts.

3.5.2 Threshold amplifier.- The requirements of 3.4.2 shall apply with the following exceptions:

(a) Diode leakage shall not exceed 0.3 microamp.
(b) Forward diode drop shall not exceed 1 volt.
(c) The true and false output shall be 6.8 $\pm$ 0.8 and 0.25 $\pm$ .25 volts respectively.

3.5.3 Input network.- The forward voltage drop shall not exceed 0.7 volts at a current of 0.9 milliamperes, and the reverse leakage shall not exceed 10 microamperes with a reverse bias of 6 volts.

3.5.4 1SR - OSR driver.- Under the input and load conditions of figure 4, driver operation shall be as follows:

3.5.4.1 True state.- With a true input applied to the base of the input transistor, the following conditions shall exist within a driver.

(a) The collector voltage of the input transistor shall be a plus 0.2 plus or minus 0.2 volts at an ambient temperature of minus 40 degrees Fahrenheit and shall be a plus 0.15 plus or minus 0.15 volts at an ambient temperature of plus 135 degrees Fahrenheit.

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(b) The voltage at the base of the output transistor shall be a minus 5.79 plus or minus 0.54 volts at an ambient temperature of minus 40 degrees Fahrenheit, and shall be a minus 5.10 plus or minus 0.54 volts at an ambient temperature of 135 degrees Fahrenheit.

(c) The collector voltage of the output transistor shall be a minus 6.42 plus or minus 0.54 volts at an ambient temperature of minus 40 degrees Fahrenheit, and shall be a minus 5.36 plus or minus 0.54 volts at an ambient temperature of 135 degrees Fehrenheit.

3.5.4.2 False state.- With a false input applied to the base of the input transistor, the following conditions shall exist within a driver.

(a) The collector voltage of the input transistor shall be minus 19.1 plus or minus 0.6 volts at an ambient temperature of minus 40 degrees Fahrenheit and shall be a minus 15.3 plus or minus 0.6 volts at plus 135 degrees Fahrenheit.

(b) The collector voltage of the output transistor shall be a plus 3.20 plus or minus 0.54 volts at ambient temperatures of minus 40 degrees Fahrenheit and shall be a plus 2.78 plus or minus 0.54 volts at plus 135 degrees Fahrenheit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

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4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 Percent Inspection)

	Characteristics	Requirements	Test Procedures
101.	Neon Driver - true state (load)	3.4.1.1	4.5.2.1.1
102.	Neon Driver - true state (no-load)	3.4.1.2	4.5.2.1.2
103.	Neon Driver - false state (no-load)	3.4.1.3	4.5.2.1.3
104.	Threshold Amplifier - input diode	3.4.2.1	4.5.2.2.1
105.	Threshold Amplifier - static condition	3.4.2.2	4.5.2.2.3
106.	Input network	3.4.3	4.5.2.3
107.	ISR-OSR Driver - true state	3.4.4.1	4.5.2.4.1
108.	ISR-OSR Driver - false state	3.4.4.2	4.5.2.4.2

4.3 Standard sampling.

4.3.1 General.- One circuit board shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

	Characteristics	Requirements	Test Procedures
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a sample from each 50 produced or from each month's production, whichever occurs first. The samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

	Characteristics	Requirements	Test Procedures
303.	Vibration	3.3.1	4.5.1.1
304.	Shock	3.3.2	4.5.1.2
305.	Humidity - Operating	3.3.4.1	4.5.1.3.2
306.	Humidity - Non-operating	3.3.4.2	4.5.1.3.1
307.	Salt spray	3.3.6	4.5.1.4
308.	Operating temperature	3.3.3.1	4.5.1.5
309.	Storage temperature	3.3.3.2	4.5.1.5

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4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

NOTE: The power supplies listed shall be variable to within plus or minus 20 percent of the nominal voltage value with a regulation no worse than 2 percent. Unless otherwise specified, the peak-to-peak ripple shall be less than 100 mv, and the output impedance shall be no greater than 2 ohms from d.c. to 500KC.

These supplies shall deliver the voltages and tolerances as specified, for room ambient tests. Unless otherwise specified, they shall be temperature compensated as shown below, and these values shall be used for all temperature extreme tests:

Nominal Volts	Minus 40°F	Plus 135°F
- 18	-19.26 ± 0.36 volts	-16.74 ± 0.36 volts
- 3	- 3.21 ± 0.06 volts	- 2.70 ± 0.06 volts
- 10	-10.7 ± 0.2 volts	- 8.3 ± 0.2 volts
- 6	- 6.42 ± 0.12 volts	- 5.58 ± 0.12 volts
+ 3	+ 3.21 ± 0.06 volts	- 2.79 ± 0.06 volts
+ 6	+ 6.42 ± 0.12 volts	+ 5.58 ± 0.12 volts
+ 35	+37.45 ± 0.7 volts	+32.55 ± 0.7 volts
+150	+160 ± 3 volts	+140 ± 3 volts

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ITEM	DESCRIPTION
1	Power supply; nominal value of 18 plus or minus 0.54 volts, current rating of at least 100 ma.
2	Power supply; nominal value of minus 3 plus or minus 0.12 volt, current requirements of at least 50 ma.
3	Power supply; nominal value of minus 10 plus or minus 0.2 volt, current rating of at least 100 ma.
4	Power supply; nominal value of minus 6 plus or minus 0.18 volts, current rating of at least 150 ma.
5	Power supply; nominal value of plus 6 plus or minus 0.12 volts, current rating of at least 100 ma.
6	Power supply; nominal value of plus 35 plus or minus 0.7 volts, current rating of at least 100 ma.
7	Power supply; nominal value of plus 3 plus or minus 0.12 volts, current rating of at least 50 milliamperes.
8	Power supply; nominal value of plus 150 plus or minus 2.32 volts, current requirement of at least 200 ma. The peak-to-peak ripple and noise shall be less than 400 mv and the output impedance shall be less than 10 ohms from dc to 500 kc.
9	Power supply; variable, from zero to minus 10 volts, current requirement of at least 50 ma. The peak-to-peak ripple and noise shall be less than 200 millivolts and the output impedance shall be less than 10 ohms from dc to 500 kc.
10	Voltmeter, D.C. Digital - A range of at least .001 to 999 volts dc, or three significant figures. Accuracy of at least ± 0.2% of reading ± 1 count. Input impedance of at least 11 megohms on all ranges.
11	Miscellaneous resistors and diodes - Refer to figures 1, 2, 3 and 4.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the circuit board rigidly mounted in its normal operating position Subsequent to the vibration (continued on the next page)

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test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the circuit board shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that received the circuit board in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The circuit board shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the circuit board shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.3 to determine compliance with the requirements of 3.5 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to determine compliance with 3.3.3.2.

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4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 neon driver circuit boards produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.

4.5.2.1 Neon driver tests.- Connect the power supplies of Table IV to their corresponding pins. Connect separate loads, identical to figure 1, to the pins of Column C, Table V. The following tests shall determine compliance with 3.4.1.

TABLE IV

POWER SUPPLIES FOR NEON DRIVER

Power Supply	Pin #
minus 3 volts	6B, 21B, 31B
minus 18 volts	18A, 30A
plus 150 volts	2A, 27A
power ground	1B, 26B

4.5.2.1.1 True state - static condition - (output loaded).

(a) Apply a true signal (minus 6 plus or minus 0.18 volts) to each of the pins of Column A, Table V. The voltage at each of the test points of Column B, shall be minus 3.7 plus or minus 0.22 volts.

(b) The voltage between the pins of Column C, and ground shall be 40 plus or minus 10 volts.

4.5.2.1.2 True state - (output not loaded).

(a) Apply a true signal (minus 6 plus or minus 0.18 volts) to each of the pins of Column A, Table V. The voltage at the pins of Column C with respect to the plus 150 volts shall be no less than 135 volts.

4.5.2.1.3 False state - static condition - (Output not loaded).

(a) Remove the minus 6 volts from every other pin of Column A and replace it with power ground. The voltage at each of the test points of Column B shall be no more negative than minus 1.1 volts.

(b) The voltage between the pins of Column C and ground shall be plus 100 plus or minus 10 volts.

(c) Interchange the voltage levels specified in (a), and make the measurements required there.

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TABLE V

Column A	Column B	Column C
7A	TP 19	2B
7B	TP 19	
8A	TP 20	4A
8B	TP 20	
9A	TP 21	3B
9B	TP 21	
10A	TP 22	5A
10B	TP 22	
11A	TP 23	4B
11B	TP 23	
12A	TP 24	6A
12B	TP 24	
13A	TP 25	5B
13B	TP 25	
17A	TP 26	23B
17B	TP 26	
16A	TP 27	22A
16B	TP 27	
15A	TP 28	22B
15B	TP 28	
14A	TP 29	21A
14B	TP 29	
31A	TP 30	33A
32A	TP 30	

4.5.2.2 Threshold amplifier tests.- The following tests shall determine compliance with 3.4.2.

4.5.2.2.1 Input logic diode.

(a) A plus 35 volt supply shall be applied to pins 23A, 35A, and 51A. Ground pins 24B, 36A and 53A. The voltage at pins 24A, 37A and 54A shall be no greater than 0.13 volts with respect to the 35 volt supply.

(b) Replace the plus 35 volts at pins 23A, 35A and 51A with minus 18 volts. With pins 24B, 36A and 53A at ground potential, the voltage at pins 24A, 37A and 54A shall be no greater than 0.7 volt with respect to ground.

4.5.2.2.2 Power and ground connections.

(a) Connect the following power supplies to the corresponding pins: plus 35 volts to pins 23A, 35A and 51A; plus 6 volts to pins 25A, 48A and 63A; minus 10 volts to pin 25B.

(b) Connect the returns of the supplies to the ground return, pin 65B.

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4.5.2.2.3 Static conditions (output and output prime loaded).

(a) Separate loads, identical to figure 2, shall be connected to each of the pins of columns C and D of Table VI. Connect the input network of figure 3 to the pins of column A.

(b) With a voltmeter connected from each of pins 24A, 37A, and 54A to ground, the input network pot shall be adjusted until the voltmeter measures minus 1.75 plus or minus 0.02 volts. Adjust the potentiometer of column B, Table VI until the voltages at the pins of column C teeter between minus 7 and 0 volts.

(c) Adjust the input network to supply minus 1.67 plus or minus 0.02 volts to column A.

(1) The voltage at the pins of column C shall be between 0 volt and minus 0.25 volts.

(2) The voltages at the pins of column D shall be minus 6.8 plus or minus 0.5 volts.

(d) Adjust the input network to supply minus 1.83 plus or minus 0.02 volts.

(1) The voltage at the pins of column C shall be minus 6.8 plus or minus 0.5 volts.

(2) The voltage at the pins of column D shall be between 0 volt and minus 0.25 volt.

TABLE VI

Co.	A	B	C	D	E	F
	IN	POT	OUT	OUT	BASE OF OUT	BASE OF OUT
	24A	R4	29A	34A	TP 6	TP 5
	37A	R6	49A	50A	TP 12	TP 11
	54A	R8	64A	65A	TP 18	TP 17

4.5.2.3 Input network.- The following tests shall determine compliance with 3.4.3.

4.5.2.3.1 Forward voltage drop.

(a) Remove all previous connections and connect a 1K, 1%, 1/4 watt resistor between pins 63A and 64B

(b) Connect a plus 6 volts to pin 63A, and connect pins 45A and 59A to power ground.

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(c) Read the voltage at pins 37B, 39B, 41B, 46B, 53B, 56B, 60B and 64B. This voltage shall be no greater than 0.6 volts with respect to power ground.

(d) Remove the ground connections from pins 45A and 59A and connect the following pins to power ground: 40A, 43A, 46A, 55A, 57A and 60A.

(e) Read the voltage at pins 39B, 41B, 46B, 53B, 56B, and 60B. This voltage shall be no greater than 0.6 volts.

(f) Remove the ground connection from pin 60A and connect pin 62A to power ground.

(g) Read the voltage at pin 60B. This voltage shall be no greater than 0.6 volts.

4.5.2.3.2 Reverse leakage current.

(a) Remove the plus 6 volts from pin 63A and then ground this pin.

(b) Remove the ground connections from pins 40A, 43A, 46A, 55A, 57A and 62A.

(c) Connect one end of a 1K, 1%, 1/4 watt resistor to power ground. Connect the other end of the resistor to each of the pins in column A of table VII, one at a time, and connect a minus 6 volts to the corresponding pin in Column B.

(d) Read the voltage at the pin in column A to which the resistor is connected. This voltage shall be no greater than 6 millivolts.

TABLE VII

A	B
45A	37B
45A	39B
45A	41B
45A	46B
40A	39B
43A	41B
46A	46B
55A	53B
59A	53B
59A	56B
59A	60B
57A	56B
60A	60B
62A	60B
61A	64B

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4.5.2.4 ISR - OSR Driver.- The following tests shall determine compliance with 3.4.4.

4.5.2.4.1 True state.

(a) Remove all previous connections and connect the following:

(1) Separate input circuits (switch open) of Fig 4 to pins 35B and 49B.
(2) Separate load resistors of fig 4 to pins 36B and 51B.
(3) Minus 6 volts to pins 48B and 63B.
(4) Plus 3 volts to pins 38A and 50B.
(5) Minus 18 volts to pin 30A.
(6) Power ground to pin 65B.

(b) Read the voltage at test points 31 and 33. This voltage shall be minus .05 plus or minus .05 volts.

(c) Read the voltage at test points 32 and 34. This voltage shall be minus 5.65 plus or minus 0.54 volts.

(d) Read the voltage at pins 36B and 51B. This voltage shall be minus 5.8 plus or minus 0.54 volts.

4.5.2.4.2 False state.

(a) Close the switch in the input circuit.

(b) Read the voltage at test points 31 and 33. This voltage shall be minus 18 plus or minus 0.6 volts.

(c) Read the voltage at pins 36B and 51B. This voltage shall be plus 2.93 plus or minus 0.55 volts.

4.5.3 Temperature extremes.- The following tests shall be performed at the test conditions of 4.5.1.3.2 and 4.5.1.5, using the temperature compensated supply voltages of 4.4.1.1 (as applicable).

4.5.3.1 Neon driver.- The following tests shall determine compliance with 3.5.1.

4.5.3.1.1 OFF state (no load).

(a) Make the supply connections listed in table IV.

(b) Apply minus 6 volts to every other pin of table V, Column A; ground the alternate pins.

(c) The voltage between the plus 150 volts and the pins of Column C shall not exceed 70 volts.

(d) Reverse the input connections of (b) and perform the test in (c).

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4.5.3.1.2 ON state (no load).

(a) Apply minus 6 volts to the pins of Column A, table V.

(b) The voltage between the plus 150 volts and the pins of Column C shall be no less than 110 volts.

4.5.3.1.3 ON state (output loaded).

(a) Apply minus 6 volts to the pins of Column A, Table V.

(b) Connect the load of figure 1 to the pins of Column C.

(c) The voltage between the plus 150 volts and the pins of Column C shall be no less than 100 volts.

4.5.3.2 Threshold amplifier.- The following tests shall determine compliance with 3.5.2.

Repeat 4.5.2.2 with the following exceptions:

(a) In 4.5.2.2.1(a) the measured voltage shall be no greater than 0.25 volts with respect to the 35 volt supply.

(b) In 4.5.2.2.1(b) the measured voltage shall be no greater than 1 volt.

(c) The measurements of 4.5.2.2.3 shall be minus 6.8 $\pm$ 0.8 volts and 0.25 $\pm$ 0.25 volts.

4.5.3.3 Input network.- Repeat 4.5.2.3 in its entirety except that the voltage read in 4.5.2.3.1(c), (e), and (g) shall be no greater than 1.2 volts, and the voltage read in 4.5.2.3.2(d) shall be no greater than 12 millivolts. These tests shall determine compliance with 3.5.3.

4.5.3.4 ISR - OSR Driver.- Repeat 4.5.2.4 in its entirety making the exceptions as noted in 3.5.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The neon driver circuit board consists of neon drivers, threshold amplifiers, and a Input Network, and ISR-OSR Driver. The intended use of each of these circuits is as follows:

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(a) Neon drivers.- The neon driver is an electronic switch, using minus 6 volt and 0 volt levels for triggering operations of neon lights. These lights provide information concerning the modes of operation of the computer.

(b) Threshold amplifier.- The threshold amplifiers provide a digital selection of some threshold voltage level which is presented in analog form due to voltage drops. It acts as a voltage selector switch to input voltage levels and converts these to true digital readout.

(c) Input network.- A clamped input circuit which reduces noise from the Memory Loading Unit (MLU) inputs. (The MLU is used with the Gun Direction Computer: M18 for programming bit information into the memory of the computer. All three pieces of equipment function as a system in conjunction with field artillery problems.)

(d) ISR - OSR Driver.- The "One Set Read" (1SR) and the "Zero Set Read" (OSR) Drivers are amplifiers used to set and reset the read flip-flops of a computer under test.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) A block of serial numbers to identify each circuit board.

Custodian:

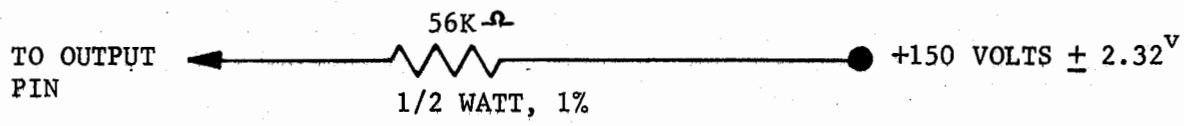
Army - MU

Preparing activity:

Army - MU

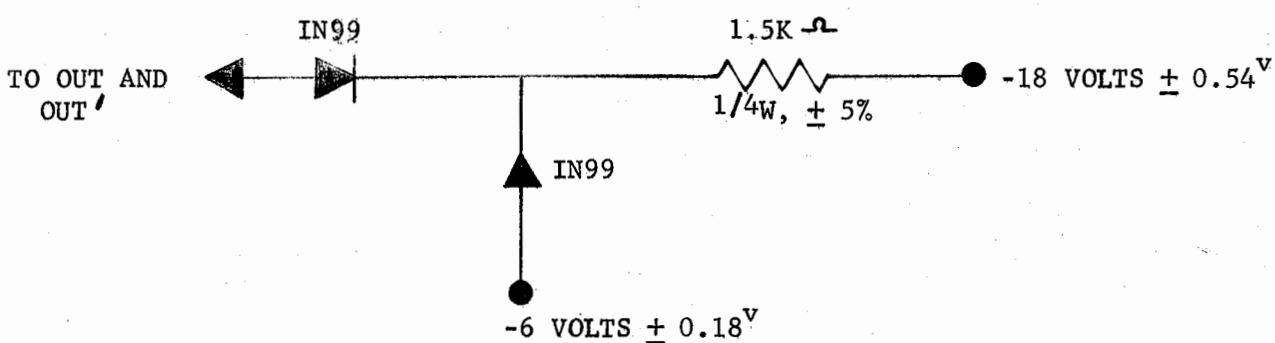
Project No. 1220-A188Y

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NEON DRIVER LOAD

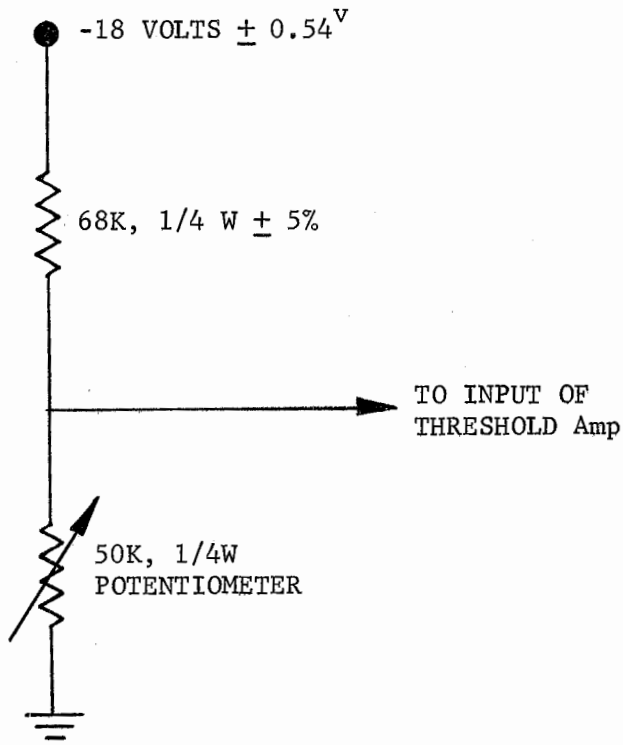
FIGURE 1



LOAD FOR THRESHOLD AMPLIFIER

FIGURE 2

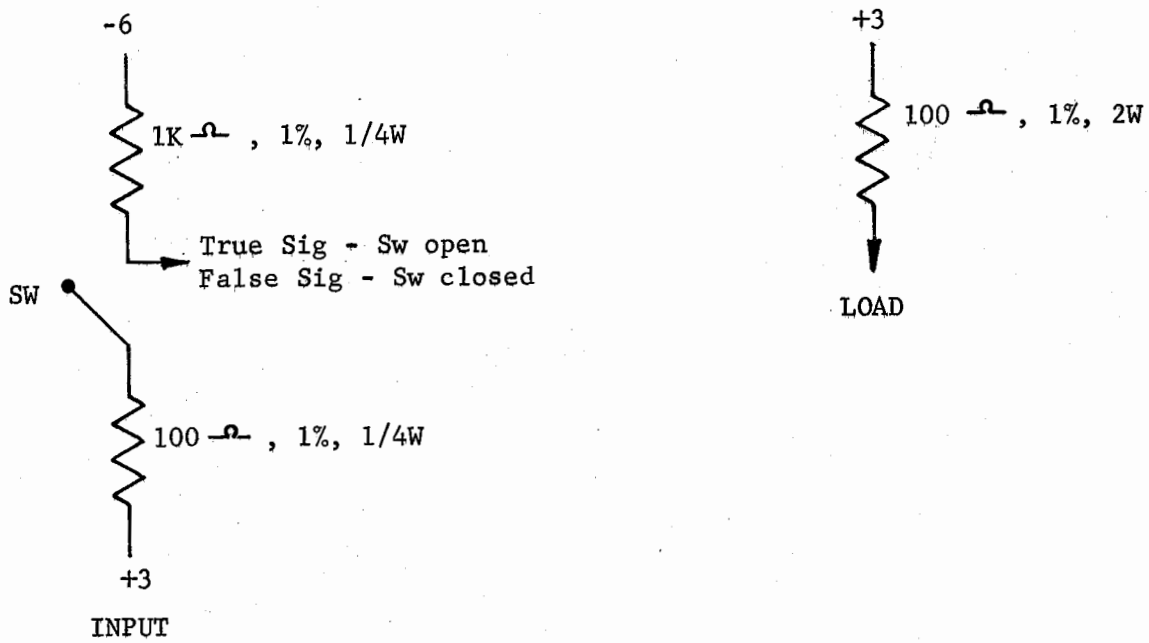
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INPUT NETWORK FOR THRESHOLD AMPLIFIER

FIGURE 3

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INPUT AND LOAD FOR 1SR-OSR DRIVERS

FIGURE 4

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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